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Characterization of Raman Scattering in Solid Samples with Different Particle Sizes and Elucidation on the Trends of Particle Size-Dependent Intensity Variations in Relation to Changes in the Sizes of Laser Illumination and Detection Area

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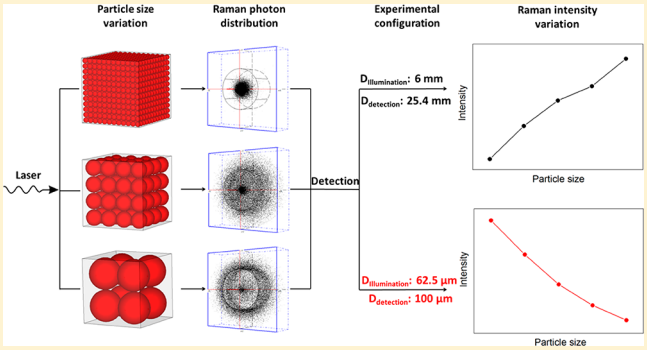
Supporting Information

ABSTRACT: We have systematically characterized Raman scatterings in solid samples with different particle sizes and investigated subsequent trends of particle size-induced intensity variations. For this purpose, both lactose powders and pellets composed of five different particle sizes were prepared. Uniquely in this study, three spectral acquisition schemes with different sizes of laser illuminations and detection windows were employed for the evaluation, since it was expected that the experimental configuration would be another factor potentially influencing the intensity of the lactose peak, along with the particle size itself. In both samples, the distribution of Raman photons became broader with the increase in particle size, as the mean free path of laser photons, the average photon travel distance between consecutive scattering locations, became longer under this situation. When the particle size was the same, the Raman photon distribution was narrower in the pellets since the individual particles were more densely packed in a given volume (the shorter mean free path). When the size of the detection window was small, the number of photons reaching the detector decreased as the photon distribution was larger. Meanwhile, a large-window detector was able to collect the widely distributed Raman photons more effectively; therefore, the trends of intensity change with the variation in particle size were dissimilar depending on the employed spectral acquisition schemes. Overall, the Monte Carlo simulation was effective at probing the photon distribution inside the samples and helped to support the experimental observations.



Raman spectroscopy has been extensively used for the analysis of diverse solid samples such as polymers, pharmaceuticals, and agricultural products.[1−9] The frequent adoption of Raman spectroscopy is attributed to the capability of providing rich chemical information about samples, as well as fast and simple data acquisition in a nondestructive manner. Meanwhile, when vibrational spectroscopic methods including Raman spectroscopy are employed for long-term robust quantitative analysis, especially combined with chemometric calibration tools,[10−14] securing the reproducibility of the measurement should be the most demanding prerequisite. As generally known, Raman spectral features of a sample could be susceptible to variation by physical parameters such as particle size, packing density, morphology, and so on. Therefore, understanding their influence on subsequent Raman spectra will be ultimately beneficial for reliable quantitative analysis.

Particle size is one of most critical issues to consider since its variation rather sensitively alters Raman spectral features. For this reason, several research groups have investigated the particle size-dependent variations of peak intensities using diverse samples. Martin et. al reported the decrease in peak intensities with the increase in particle sizes, when $LiCO_3$, NH_4Cl, $Ba(NO_3)_2$, and K_2CrO_4 powders were measured with FT-Raman instrumentation.[15] The authors explained that the smaller particles led to the increase in the laterally spread near-surface Raman signal, and the more efficient gathering of Raman photons using a collection optical fiber was the origin for the increased intensity for the samples composed of smaller particles. The same trend was observed in the measurements of $NaNO_3$, $NaClO_3$, $NaBrO_3$, $K_4Fe(CN)_6{\cdot}3H_2O$, $CuSO_4{\cdot}5H_2O$, and $K_3Fe(CN)_6$ powders.[16] Also, Hu et. al found the same trend through the examination of the Raman spectra of flufenamic acid samples with different particle sizes.[17] Meanwhile, the observed trend in these publications was opposite to the expectation based on Kubelka−Munk theory.[18] The theory predicts the diminishment of Raman signals with the decrease in particle size due to the decreased sampling volume, since the

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decreased particle size causes the diffusion of laser photons in a relatively small volume.

In contrast to the above reports, a dissimilar trend was found in the measurements of aspirin, Avicel, and $NaNO_3$, in which both the laser illumination (diameter: 6 mm) and photon-acceptance area (diameter: 25.4 mm) were large.[19] Their peak intensities elevated with the particle size increase of up to 275 μm for Avicel as well as aspirin and 400 μm for $NaNO_3$ and then decreased afterward. As explained by the authors, the particles larger than the mentioned sizes made the photons much more laterally spread over the sample, and some portion of the photons was not able to reach to the detector, so the decrease in peak intensity was the result.

The trends of particle size-dependent spectral variations in the previous research were generally similar; however, the discordant observation was also reported as described. It originated from the fact that the physical properties of the tested samples, such as refractive index and crystallinity, were dissimilar, and/or the instrumental configurations for spectral acquisition, such as size of laser illumination, size of detection window, and photon collection efficiency, were different. Therefore, the found trends were on a rather case-by-case basis and probably difficult to generalize.

This study aims to conduct a systematic investigation on the particle size-dependent Raman spectral features to potentially explain the origin of the discordant observations. For this purpose, both pure lactose powders and compressed lactose pellets composed of five different particle sizes were selected, and three different spectral acquisition schemes (refer to Figure 1) were also uniquely employed to collect Raman spectra of the

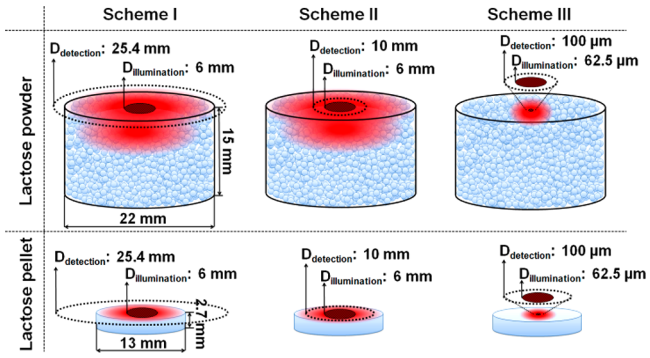

Figure 1. Schematic illustration for the laser illumination and photon detection geometries of Scheme I, II, and III.

samples. The adoption of the pellet was to mimic a highly compressed sample such as a pharmaceutical tablet. Initially, backscattering Raman spectra of the samples were collected using an instrument providing a large illumination area (diameter: 6.0 mm) and a wide detection window (diameter: 25.4 mm). Next, only the diameter of the detection window was narrowed to 10.0 mm, and Raman spectra were collected again. Finally, a separate instrument possessing an excitation diameter of 62.5 μm and a collection diameter of 100 μm was employed for the same measurement. The trends of the particle size-induced intensity variations of the two different sample types in relation with the employed spectral acquisition schemes were analyzed.

The Monte Carlo simulation has been known as a powerful method for studying photon migration in diverse samples, such as for predicting how Raman intensities of subsurfaces or multilayered tissue objects with various shapes vary for a range of experimental variables.[20−25] In this study, Monte Carlo simulation was also used to visualize distributions of Raman photons in the samples, counting the numbers of Raman photons reaching a detector in all cases. Finally, the intensity variations observed in the experimental data were compared with those obtained from the simulation.

■ EXPERIMENTAL SECTION

Sample Preparation and Acquisition of Raman Spectra. Pure lactose (β-lactose) powder was purchased from Sigma-Aldrich. Lactose powders with five different particle sizes (38 ± 26, 87 ± 36, 157 ± 87, 261 ± 119, and 382 ± 122 μm) were prepared by sieving the purchased sample with the proper sizes of sieves. The particle sizes were determined using a particle size analyzer (Malvern Instruments, Malvern, U.K.). The lactose pellets were made by placing 500 mg of each powder in a stainless-steel die (diameter: 13 mm) and pressing it at a constant pressure of 5 ton/cm^2 for 2 min.

Raman spectra of the samples were collected by a back-scattering mode with three different schemes as schematically presented in Figure 1. Initially, a spectrometer (RamanRxn1 unit, Kaiser Optical Systems, Ann Arbor, MI) equipped with a PhAT probe providing a large illumination area (28.3 mm^2; diameter: 6 mm) and a wide photon-collection area (506.7 mm^2; diameter: 25.4 mm) was used (Scheme I). Powder samples were added into a glass vial (diameter: 22 mm), and the packing thickness was maintained at approximately 15 mm for each measurement. The bottom of the glass vial contained with the sample was illuminated for spectral acquisition. For the measurement of pellets, each pellet was directly positioned at the focal point for spectral acquisition. Next, the Raman spectra of the same samples were also acquired by decreasing the diameter of the detection window to 10.0 mm, while maintaining the size of illumination unchanged (Scheme II). For this purpose, a metal block possessing a 10.0 mm hole at the center was placed in front of the PhAT probe for collection of spectra. Finally, a separate Raman probe (Kaiser Optical Systems) providing an excitation diameter of 62.5 μm and a collection diameter of 100 μm was also employed to measure the same samples (Scheme III). The laser wavelength was 785 nm, and the resolution of all collected spectra was 4 cm^{-1}. Baseline correction of the spectra was performed using MATLAB version 7.0 (MathWorks Inc., MA).

Monte Carlo Simulation. For the simulation, it is necessary to define the absorption coefficient (μ_a) and scattering coefficient (μ_s), corresponding to the probability of photons being absorbed and probability of photons being scattered, respectively, as below

$$\mu_a = \rho\sigma_a$$

$$\mu_s = \rho\sigma_s$$

where σ_a and σ_s correspond to the cross-sectional areas of absorption and scattering particles, respectively. ρ is the number of relevant particles in a sample medium. It is assumed that μ_a and μ_s are uniform throughout the medium. The units of μ_a and μ_s are cm^{-1}. The absorption coefficient corresponds to the degree of overall intensity attenuation resulted from not only absorption but also diverse causes. The reciprocal of μ_a is the absorption mean free path, corresponding to the average photon travel distance between initial and next absorption locations. Similarly, the reciprocal of μ_s is the scattering mean

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Table 1. Average Lactose Particle Sizes and Their Physical Parameters Corresponding to Both 785 (Laser Wavelength) and 808 nm Used in the Monte Carlo Simulation[a]

		785 nm				808 nm			
		absorption coefficient, μ_a (cm^{-1})		reduced scattering coefficient, μ'_s (cm^{-1})		absorption coefficient, μ_a (cm^{-1})		reduced scattering coefficient, μ'_s (cm^{-1})	
average particle size (μm)	refractive index	powder	pellet	powder	pellet	powder	pellet	powder	pellet
38	1.652	0.09	0.19	681.0	1475.7	0.009	0.19	642.0	1391.2
87				280.9	608.6			277.9	602.1
157				146.5	315.2			154.6	335.1
261				77.6	168.2			92.9	201.3
382				42.1	97.8			63.3	137.2

[a]The 808 nm is the equivalent wavelength to the strongest lactose peak at 359 cm^{-1} as shown in Figure 2.

free path, corresponding to the average travel distance of a photon between consecutive scattering points.

Since scatterings occur multiply upon interaction with a sample, the diffusion approximation is commonly used to simplify the complex events. So, the reduced scattering coefficient (μ_s') is commonly employed as shown below:

$$\mu_s' = \mu_s(1 - g)$$

where g is the anisotropy factor. A nonzero g value is representative of anisotropic scattering. A g value of 1 corresponds to total forward scattering. As reported previously,[26] considerable forward scattering occurred inside solid pharmaceutical samples, since the size of the particle was much greater than the wavelength of the laser, such as the 785 nm in this study. The parameters corresponding to both 785 and 808 nm used in the simulation for each sample (refractive index, μ_a, and μ_s') are summarized in Table 1. The 808 nm is the equivalent wavelength to the strongest lactose peak at 359 cm^{-1} as shown in Figure 2. These were obtained from the previous work by Zhigang et al.[27]

The Monte Carlo simulation package developed by F. F. De Mul[28,29] was used. Initially, a location of laser irradiation (λ_1) in a medium was determined. At this location, either absorption of laser photons, Rayleigh or Raman scatterings, could occur. Then, the Raman scattering photons generating from the above position with wavelength λ_2 (longer than λ_1, Stokes Raman emission) were tracked in the simulation. The number of illuminated laser photons was 100 000, and the laser photons were homogeneously distributed over the illumination area. The diameters of the illumination area and detection window inputted for the simulation were equal to the actual sizes of the experimental settings.

RESULTS AND DISCUSSION

Raman Spectra of Lactose Powders with Different Particle Sizes. Figure 2a shows Raman spectra of pure lactose powders with five different particle sizes (designated as PO-1 through PO-5) in the 1080−321 cm^{-1} range collected using Scheme I (6 mm illumination and 25.4 mm detection window). The most noticeable observation among the spectra is that the baselines of the spectra move upward vertically as the particle size of the lactose increases. The lactose peaks are present on top of the corresponding underlying baselines. To investigate the intensity variation of the lactose peaks, the baselines of the spectra were linearly corrected at seven different wavenumbers (1080, 1025, 1002, 971, 863, 805, 500, 321 cm^{-1}) as shown in Figure 2b. The inset highlights the lactose peaks in the 500−321 cm^{-1} range, where the most intense peak at 359 cm^{-1} is

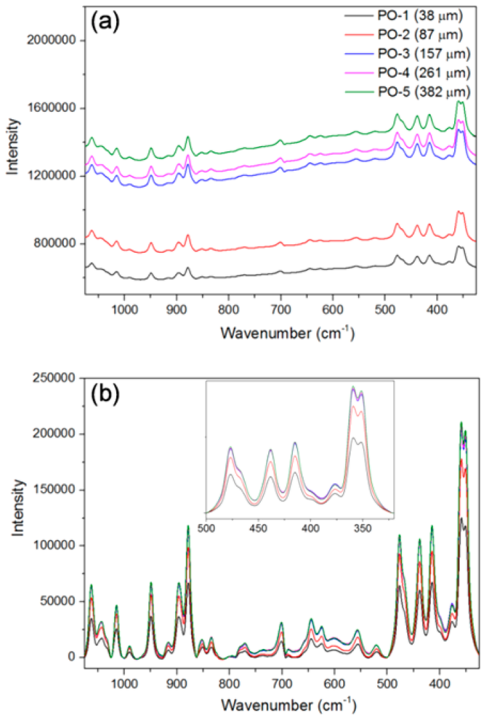

Figure 2. Raman spectra of lactose powders composed of particles with five different sizes in the 1080−321 cm^{-1} range (a) and the corresponding baseline-corrected spectra (b). The inset highlights the lactose peaks in the 500−321 cm^{-1} range. The spectra were collected using Scheme I (6 mm illumination, 25.4 mm detection window).

located. The intensities of the peaks rise as the size of the lactose particle becomes larger. Figure 3a displays the variation of the peak intensity at 359 cm^{-1} (open black circle) according to the particle size. The intensity clearly elevates with the increase in particle size, and the elevation becomes less steep when the particle sizes are larger.

To explain the observed trend above, opposite to those found in the most of the previous studies[15−17] listed in the introduction, a Monte Carlo simulation was employed to examine photon propagation in the lactose powders. Figure 4 shows simulated three-dimensional distributions of generated Raman photons in the lactose powders composed of particles with five different sizes. The dots indicate the spatial locations of the Raman photons. In the case of smallest particle size (PO-1), the photons are densely populated near the laser illumination spot. When a particle is small, the scattering coefficient of 785 nm radiation is large, as shown in Table 1, or equivalent to a small mean free path, since the reciprocal of the

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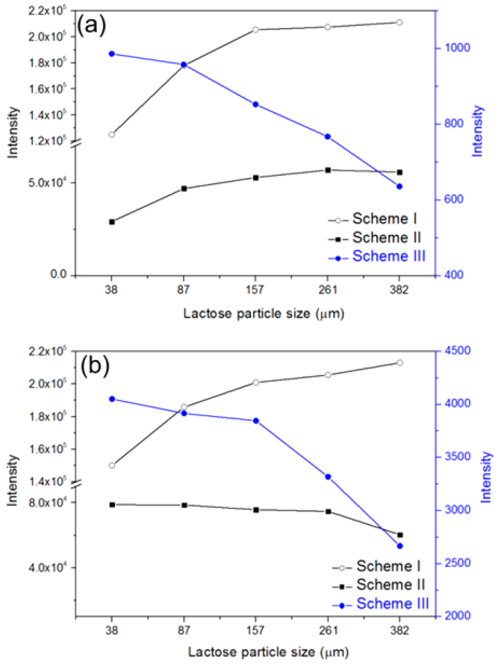

Figure 3. Variation of the peak intensity at 359 cm^{-1} in the measurements of lactose powders (a) and pellets (b) composed of particles with five different sizes when Schemes I (open circle), II (back square), and III (blue circle) are separately employed for spectral acquisition. In the case of Scheme III, refer to the right *y*-axis.

scattering coefficient corresponds to the mean free path. The small mean free path implies that the average photon travel distance between consecutive scattering locations is short. Therefore, the diffusion of laser radiation occurs in a relatively small volume, and the distribution of the Raman photons is subsequently more populated around the laser illumination area. The result is in accordance with Kubelka−Munk theory, which predicts the diffusion of photons in a relatively localized space when the sample particles are small, as explained in the introduction. In the case of PO-2, the photon distribution inside the sample becomes broader and deeper. The mean free path becomes longer in PO-2 from the decreased scattering coefficient (refer to Table 1), so the laser photons are able to diffuse farther in the sample. The photon distribution becomes substantially greater with the further increase in particle size, and the photons are extensively spread in the case of PO-5.

For more detailed characterization, Raman photon distributions along the depth as well as over the lateral direction were examined by the Monte Carlo simulation. Figure 5a shows the number of photons distributed along the depth in the lactose powders. In the case of PO-1, the photons are mostly populated just beneath the sample surface, from 0 to approximately 3.0 mm in depth. With increasing particle size, the location of maximum photon population moves deeper inside the sample, and the distribution becomes substantially broader. This tendency was also observed in the previous report by Dieter et al.[25] Figure 5b shows the number of Raman photons along the lateral direction (diameter line) on top of packed lactose powders in a round vial. The data are arbitrarily offset for easier comparison. Zero corresponds to the center. The photons are very closely located around the center in the case of PO-1 and become widely spread from the center to both sides with the increase in particle size.

In detail, the center dips are observed and the populations of Raman photons become increased at both edges in all cases. Upon the laser irradiation on the center, the generated Raman photons immediately propagate toward both sides as well as the bottom. Therefore, the population of Raman photons at the center ($x = 0$) could be substantially small, thereby explaining the center dip. The edge of powder packing corresponds to the interface between lactose and air. When generated internal Raman photons propagate toward the edge with a random-walk-like fashion, some of them approaching smaller than the critical angle are reflected inside the packing, since the refractive index of lactose is higher than that of air. Therefore, the population of photons near both edges becomes greater due to the internal reflection. Overall, the observations in Figures 4 and 5 confirm that the larger particles make the distribution of Raman photons deeper and wider inside the samples.

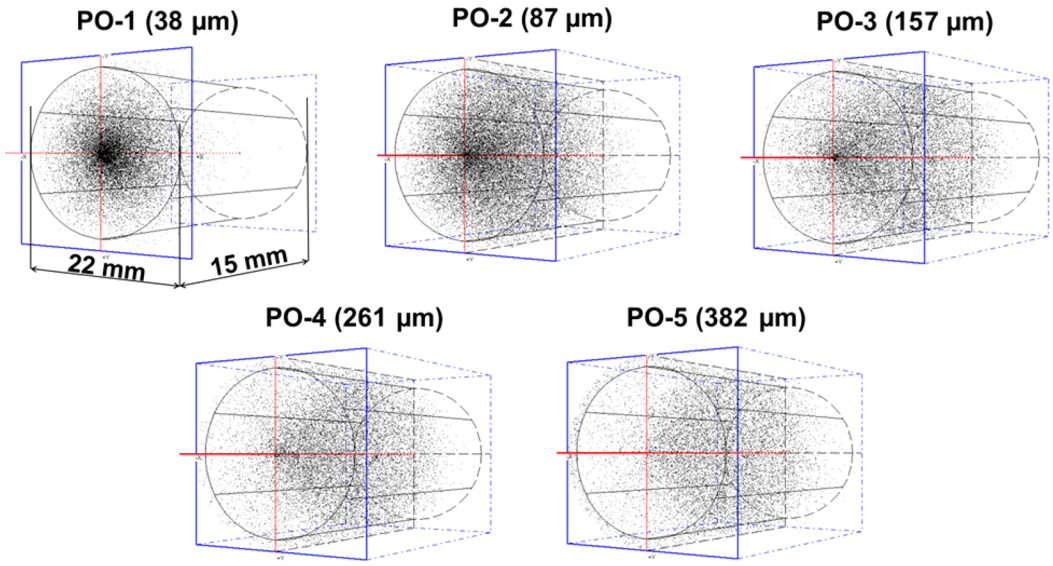

Figure 4. Simulated three-dimensional distributions of Raman photons in the lactose powders composed of particles with five different sizes. Laser is irradiated at the center of the left circle (top of lactose packing), and the dots indicate the spatial locations of the Raman photons. The dimensions of the powder packing are also shown.

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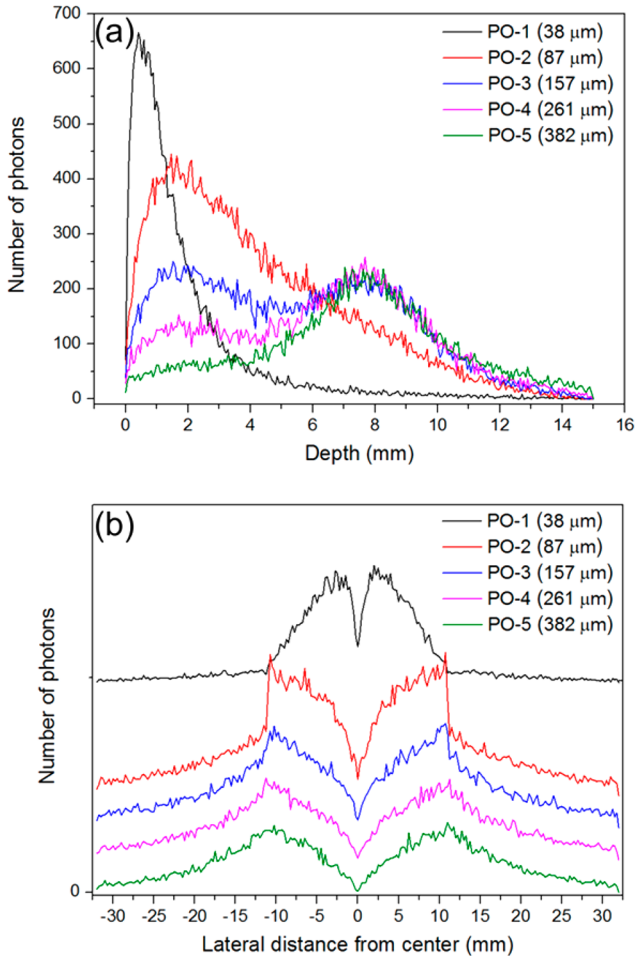

Figure 5. Simulation results on the number of photons distributed along the depth (a) and the lateral direction on top of lactose powders (b) composed of particles with five different sizes. The lateral distributions are arbitrarily offset for clear comparison.

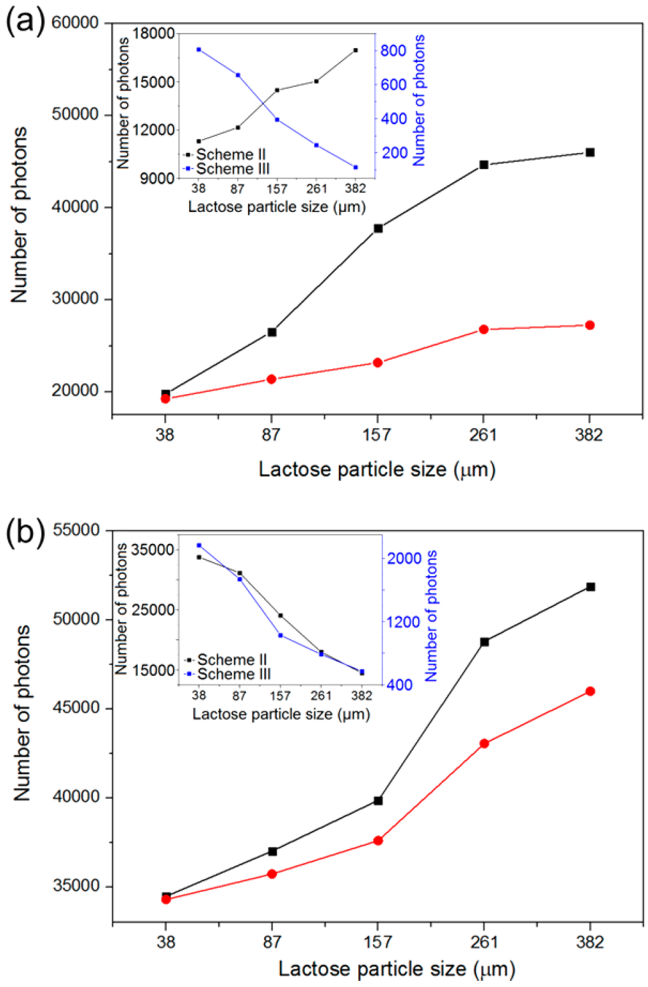

Figure 6. Simulation results on the numbers of total backscattered Raman photons (black square) and Raman photons counted by the detector (red circle) in the measurements of lactose powders (a) and pellets (b) composed of particles with five different sizes using Scheme I. Under the same situation, simulation results on the numbers of detected Raman photons in the cases of using Schemes II and III are also shown in the respective insets.

Since the distributions of photons inside the samples varied depending on the particle size, the number of photons arriving at a given detecting area is expected to change accordingly. For the evaluation, the numbers of Raman photons reaching the detector (red circles, backscattering measurement) were calculated for each sample using the Monte Carlo simulation as shown in Figure 6a. Also, the numbers of total backscattered photons (black squares), counted by assuming the use of a detection window with an infinite size, are plotted together. The number of total backscattered photons increases as the particle size becomes larger due to the greater diffusion of photons. The number of detected (counted) photons also increases, which is accordant with the experimental observations (refer to Figure 3a, Scheme I). Since the detection window here was considerably wide (larger than the diameter of powder packing), it was still able to collect a considerable portion of the widely dispersed backscattered Raman photons, such as the collection percentage of 97.3% in the case of PO-1. The collection percentage decreases as the sample particle size becomes larger due to the increase of Raman photon distribution, thereby resulting in 80.5, 61.3, 59.9, and 54.2%, in the cases of PO-2, PO-3, PO-4, and PO-5, respectively.

The above simulation aroused interest about the potential variation of the acquired Raman intensity when the size of the detection window changed. Therefore, the diameter of the detection window was decreased to 10 mm while maintaining the illumination diameter unchanged for the measurements (Scheme II), and further, Scheme III incorporating an excitation diameter of 62.5 μm and collection diameter of 100 μm (referred to as Scheme III) was separately employed as described earlier. Figure 3a shows the peak intensities of the same lactose powders, experimentally acquired using Schemes II and III. In the case of Scheme II, the overall peak intensities are lower than those of Scheme I due to the decrease in the number of detected photons from the narrowed detection window. Nonetheless, the intensity still elevates with the increase in particle size; however, the intensity of PO-5 is minutely diminished compared with that of PO-4. Since the photon distribution in PO-5 was greatest, a relatively larger portion of photons was unable to arrive at the detector when the size of the detection window was smaller, and subsequently a decrease in Raman intensity occurred. In parallel, the simulation also predicted the same trend (refer to the inset in Figure 6a).

When Scheme III was adopted, the peak intensity instead decreased as the particle size increased as shown in Figure 3a

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(refer to the scale of right *y*-axis), accordant with those found in the previous studies.[15−17] In the case of PO-1, since the size of photon distribution was nearly similar to that of employed detecting window, a great portion of the generated Raman photons was gathered at the detector. From PO-2 to PO-5, the corresponding photon distributions became greater than the size of the detection window, so less photon was able to arrive at the detector. The simulation also predicted the same trend (also refer to the inset in Figure 6a).

Raman Spectra of Lactose Pellets Composed of Particles with Different Sizes. To mimic a pressed (condensed) solid sample such as a pharmaceutical tablet, the lactose pellets (diameter: 13.0 mm, thickness: 2.7 mm approximately) were prepared from the same powders using an IR presser (designated as PE-1 through PE-5). Figure S1a shows Raman spectra of these pellets collected using Scheme I. Although the magnitude of vertical baseline variations is smaller than that observed in the measurement of powders, the baseline offset also becomes greater with the increase of lactose particle size. Similarly, the peak intensity at 359 cm^{-1} increases as the particle size becomes larger, as also observed in the measurements of powders. Meanwhile, individual lactose particles are more densely packed with much less void volume in the pellet, and the narrower distribution of Raman photons is expected because laser photons are able to diffuse only in a small volume. As estimated by the Monte Carlo simulation, the photon distribution in the pellet is considerably smaller (refer to Figure S2) in comparison with that in the powder composed of the same-size lactose particle.

Figure 7a and b shows the numbers of photons distributed along the depth and the diagonal line on top of the pellets, respectively, calculated by the simulation. The patterns are generally analogous to those observed for the powders; the depths of photon distribution are substantially shallow as expected, and the photons are more densely gathered around the illumination center and their distributions are narrower compared with those in the powders. Meanwhile, with the increase in particle size, the photons also spread to both sides from the center.

The numbers of total backscattered photons and detected photons in the measurements of pellets calculated from the simulation are shown in Figure 6b. The number of detected photons also elevated with the increase in particle size, agreeing with the trend in the experimental observation (Figure 3b), since the detecting window was large enough to recognize the backscattered photons which had the narrower distributions. The photon collection percentages in the cases of PE-1, PE-2, PE-3, PE-4, and PE-5 were 99.5, 96.5, 94.3, 88.2., and 84.6%, respectively.

The same lactose pellets were measured again using Schemes II and III, and the resulting intensity variations of the 359 cm^{-1} peak were investigated as shown in Figure 3b. Interestingly, when Scheme II was employed, the intensity decreased with the increase in particle size, reversed compared to the trend in the measurement of powders. In PE-1, since the photons were densely localized between −5 and 5 mm in the lateral direction based on the simulation (refer to Figure 7b), most of the generated photons were able to reach the detector. From PE-2 to PE-5, the photon distribution became wider and deeper; therefore, some of the photons positioned at the rim of the distribution became unable to reach to the detector, and so the intensity decreased accordingly. When Scheme III was employed, the decrease in intensity with the increase in particle size was observed (Figure 3b), as also found in the measurement of lactose powders. Again, the simulations predicted the accordant trends of intensity variations to those found in the measurements using Schemes II and III (refer to the inset in Figure 6b).

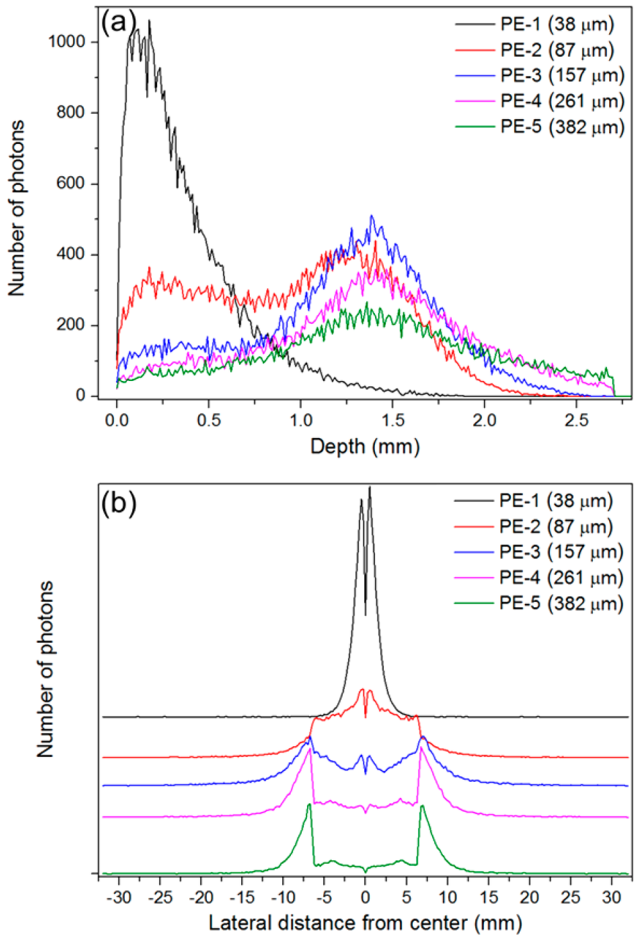

Figure 7. Simulation results on the number of photons distributed along the depth (a) and the lateral direction on top of lactose pellets (b) composed of particles with five different sizes. The lateral distributions are arbitrarily offset for clear comparison.

The overall observation in this study implies that the trends of the particle size-induced intensity variations are not always consistent and are simultaneously influenced by several factors. The physical characteristics of a sample which determine the distribution of Raman photons are the major factors. This includes diverse parameters such as particle size, refractive index, packing density, and others. The experimental configuration, such as the size of the detection window influencing the photon collection percentage, should be another factor to consider critically; however, it has not been systematically investigated so far. Now, it becomes clear that the dissimilar trends of intensity variations due to the change in particle size reported in the previous publications[15−17,19] were due to the differences in the physical properties of the tested samples as well as the employed experimental setups in each study.

■ CONCLUSION

We have systematically investigated the particle size-induced Raman spectral features by varying the physical conditions of

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the lactose sample as well as the spectral acquisition schemes. The distribution of Raman photons in the sample and subsequent experimental configuration for photon collection were the major factors simultaneously governing the final peak intensity of the sample. Therefore, when variation of the Raman intensity by the particle size of sample is discussed, a careful comparative evaluation is required since it is influenced by not only particle size but also the employed experimental configuration. This finding could make analysis much more contemplated when the reliable quantitative analysis of solid samples using Raman spectroscopy is intended. In final, Monte Carlo simulations were effective for probing the photon distribution inside the samples and helpful for supporting the experimental observations. Future study will be directed toward performing a similar evaluation employing more diverse samples with different particle shapes and crystallites, and demonstrating rather a generalized trend of intensity variations in relation to the corresponding physical characteristics of the samples.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.7b01400.

Raman spectra of lactose pellets (Figure S1) and simulated three-dimensional distributions of Raman photons in the lactose pellets (Figure S2) (PDF)

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Notes

The authors declare no competing financial interest.

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